EE2002 Tutorial 8 - Solutions

1.Consider the circuit shown in Figure 1. Identify which FSM (Mealy/Moore) it represents. Also, construct the state table.

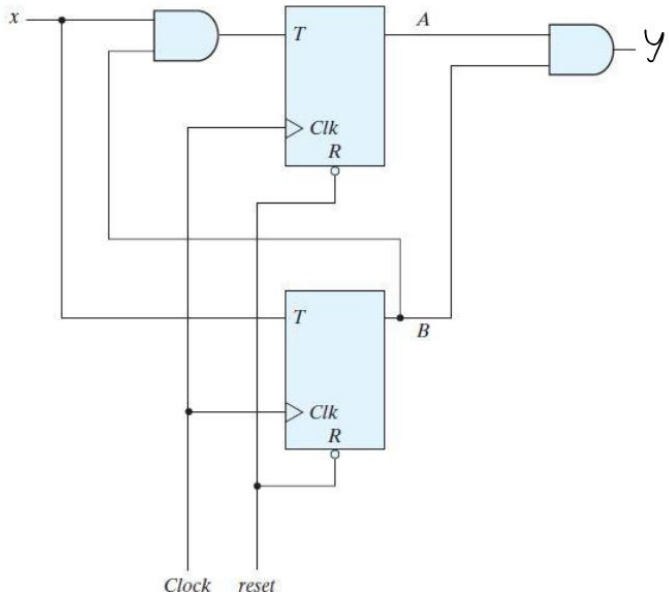


Figure 1: Sequential logic circuit

Solution:

The circuit shown represents a Moore finite state machine. The output depends only on the present state and is independent of the input.

The characteristic equation of T flip-flop is:

$$Q(t+1) = TQ' + T'Q$$

Now consider the sequential logic circuit in the question. It has two flip-flops A and B, one input x, and one output y, and can be described algebraically by the following equations:

$$T_A = Bx, \quad T_B = x, \quad y = AB$$

The values for y are obtained from the output equation. The values for the next state can be derived from the state equations by substituting T_A and T_B into the characteristic equations, yielding:

$$A(t+1) = (Bx)'A + (Bx)A' = AB' + Ax' + A'Bx$$
$$B(t+1) = x \oplus B$$

The state table for the circuit is given in Table 1. The next-state values for A and B are obtained from the two state equations.

Present State		**Input**	**Next State**		**Output**
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

Table 1: State Table

2.Given below are the state table and state assignment for a synchronous sequential circuit. Realize the circuit using D flip-flops.

$Q_2Q_1Q_0$	Present State	Next State, Output (for input X = 0)	Next State, Output (for input X = 1)
0 0 0	A	B,0	E,0
0 0 1	B	A,1	C,1
0 1 0	C	B,0	C,1
0 1 1	D	C,0	E,0
1 0 0	E	D,1	A,0

Solution:

We are given a state table with state assignments and an input X, and we need to realize the circuit using D flip-flops. The state transitions are defined for different values of X.

We assign binary strings to states A, B, C, D, and E as follows:

$$A = 000, \quad B = 001, \quad C = 010, \quad D = 011, \quad E = 100$$

We will use 3 flip-flops. To determine the next state logic for each of the flip-flops, we need to analyze the transitions in the state table for each value of X.

We create a table for Q_2^+, Q_1^+, Q_0^+ (next state) and the corresponding D flip-flop inputs D_2, D_1, D_0 for each combination of present state (denoted by $Q_2Q_1Q_0$) and X. We will denote the output by Y.

The state transition table and the K-maps are shown.

XQ2Q1Q0	**Q2+Q1+Q0+**	**D2D1D0**	**Y**
0000	001	001	0
0001	000	000	1
0010	001	001	0
0011	010	010	0
0100	011	011	1
0101	xxx	xxx	x
0110	xxx	xxx	x
0111	xxx	xxx	x
1000	100	100	0
1001	010	010	1
1010	010	010	1
1011	100	100	0
1100	000	000	0
1101	xxx	xxx	x
1110	xxx	xxx	x
1111	xxx	xxx	x

Table 2: State transition table



XQ2 \ Q1Q0	00	01	11	10
00	0	0	0	0
01	0	X	X	X
11	0	X	X	X
10	1	0	1	0

Result

$F = X\,Q1\,Q0 + X\,\overline{Q2}\,\overline{Q1}\,\overline{Q0}$

(a) Solution for D2

XQ2 \ Q1Q0	00	01	11	10
00	0	0	1	0
01	1	X	X	X
11	0	X	X	X
10	0	1	0	1

Result

$F = \overline{X}\,Q2 + X\,Q1\,\overline{Q0} + X\,\overline{Q1}\,Q0 + \overline{X}\,Q1\,Q0$

(b) Solution for D1

XQ2 \ Q1Q0	00	01	11	10
00	1	0	0	1
01	1	X	X	X
11	0	X	X	X
10	0	0	0	0

Result

$F = \overline{X}\,\overline{Q0}$

(c) Solution for D0

XQ2 \ Q1Q0	00	01	11	10
00	0	1	0	0
01	1	X	X	X
11	0	X	X	X
10	0	1	0	1

Result

$F = \overline{X}\,Q2 + \overline{Q1}\,Q0 + X\,Q1\,\overline{Q0}$

(d) Solution for Y

Figure 2: K-Map solutions for D2, D1, D0, and Y.

3.Derive the state table and state diagram for the synchronous ciruit shown in Figure 3:

Solution:

The Boolean equations for the system are as follows:

$$
\begin{aligned}
Y &= Q_1^+ Q_2^+ \\
D_2 &= Q_2 C' \oplus Q_2 C Q_1' \oplus Q_1 \cdot C Q_2' \\
D_1 &= Q_1 \oplus C
\end{aligned}
$$

Where $\oplus$ denotes the XOR (exclusive OR) operation.

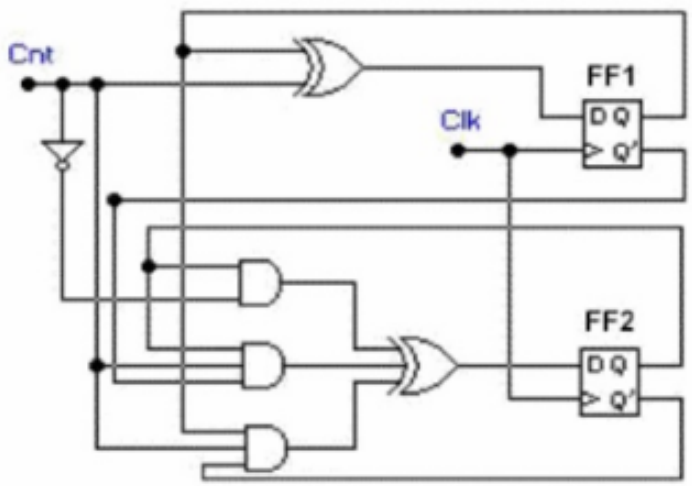


Figure 3: Synchronous circuit

Cases for $C = 0$ and $C = 1$

When $C = 0$, the values of D_2 and D_1 simplify as follows:

$$D_2 = Q_2$$
$$D_1 = Q_1$$

When $C = 1$, the values of D_2 and D_1 simplify as:

$$D_2 = Q_1'Q_2 + Q_2'Q_1$$
$$D_1 = Q_1'$$

C	Q_2	Q_1	D_2	D_1	Q_2^+	Q_1^+	Y
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	1	0	1	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	0	1	0
1	0	1	1	0	1	0	0
1	1	0	1	1	1	1	1
1	1	1	0	0	0	0	0

Table 3: State Transition Table

The state diagram is shown below figure 4

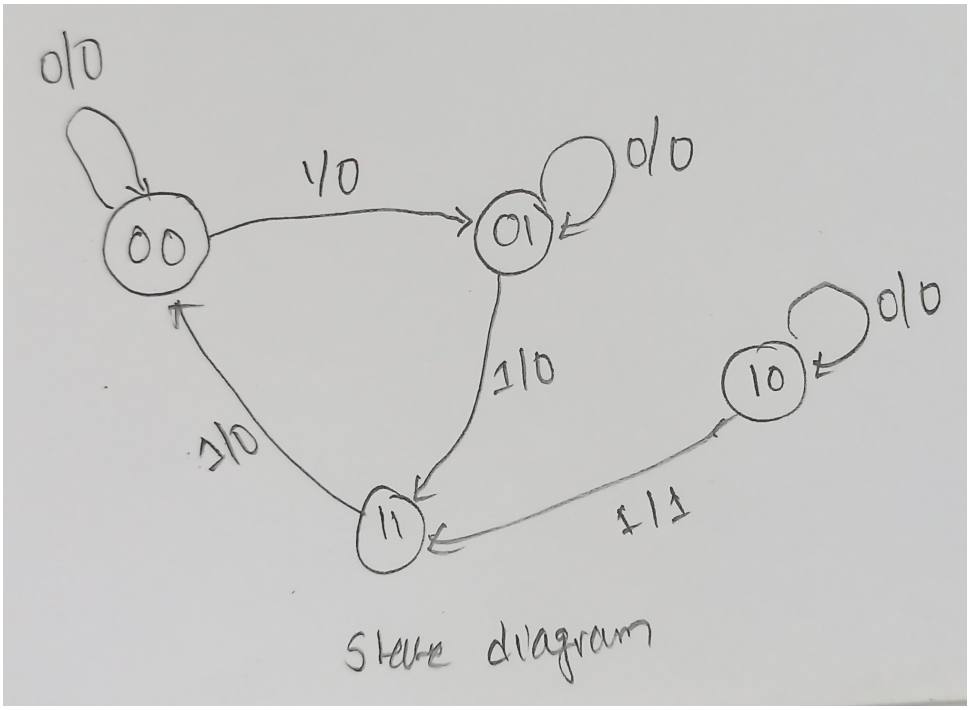


Figure 4: State diagram

4. A sequence generator is shown in Figure5. The counter state ($Q_0Q_1Q_2$) is initialized to 010 using the preset/clear inputs. Determine the sequence generated (until it repeats).

Solution:

From the given diagram, the Boolean equations for the next state inputs are:

$$D_0 = Q_2'$$
$$D_1 = Q_0'$$
$$D_2 = Q_0Q_1'$$

State Sequence

The given state sequence follows the transitions:

$$010 \rightarrow 110 \rightarrow 100 \rightarrow 101 \rightarrow 001$$

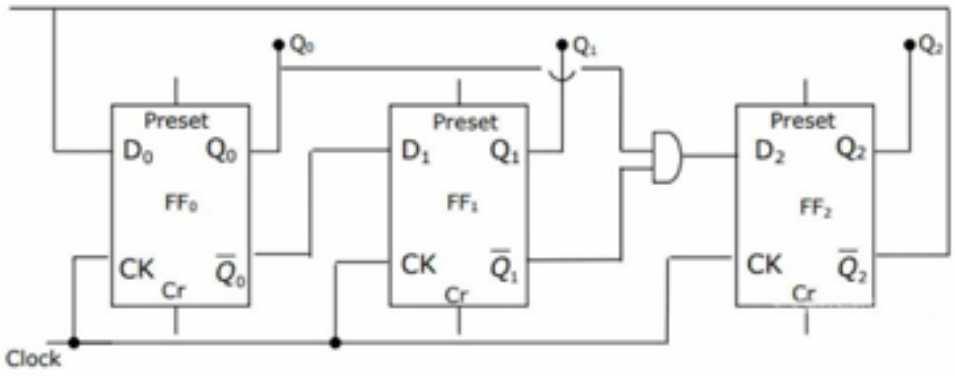


Figure 5: Sequence Generator

Q_0	Q_1	Q_2	D_0	D_1	D_2	Q_0^+	Q_1^+	Q_2^+
0	0	0	1	1	0	1	1	0
0	0	1	0	1	0	0	1	0
0	1	0	1	1	0	1	1	0
0	1	1	0	1	0	0	1	0
1	0	0	0	1	0	1	0	1
1	0	1	1	0	1	0	0	1
1	1	0	0	0	1	1	0	0
1	1	1	1	0	0	0	0	0

Table 4: State Transition Table

5. Consider the state table shown in Figure 6

(a) Draw the corresponding state diagram.

(b) Tabulate the reduced state table.

(c) Draw the state diagram corresponding to the reduced state table.

Solution:

The State diagram for the given state table (non- reduce state table) as shown in figure 7

The reduced state table and reduced state diagram are shown in Figures 9 and 9

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

Figure 6: State table

6. Realize the state table shown using the minimum number of D flip-flops.

P.S.	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	e	c	0	1
c	a	d	0	1
d	e	f	0	1
e	a	f	0	0
f	g	d	0	1
g	a	b	0	0

Solution:

Without reduction: there are 7 states
So we need 3 flip-flops.

With reduction:

From the table,
The outputs and next states of a and g are equal.
$\Rightarrow$ States a and g are equivalent.
$\Rightarrow$ We can replace $a = g$.
Similarly, row 3 = row 6 $\Rightarrow$ $c = f$ are equivalent.
After replacement,

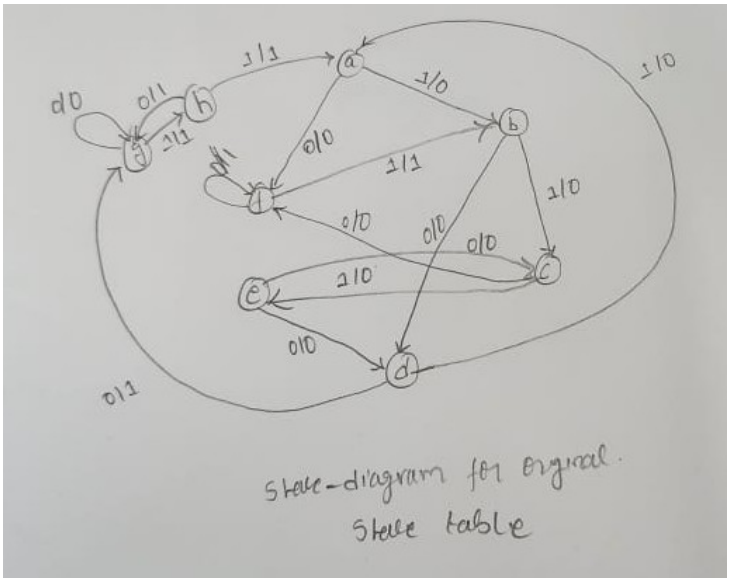


Figure 7: Orginal state table ,state diagram

Present state	Next state 0	Next state 1	Output 0	Output 1
a	f	b	0	0
b	d	a	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1

Figure 8: Reduced State table

$\Rightarrow$ Row 2 = Row 4:
$c = f = g = a$
$b = d$

Reduced Table

P.S.	State Assignment	n.s $x=0$	n.s $x=1$	Output $x=0$	Output $x=1$
00	$= a$	a	b	0	0
01	$= b$	e	c	0	1
10	$= c$	a	b	0	1
11	$= e$	a	c	0	0

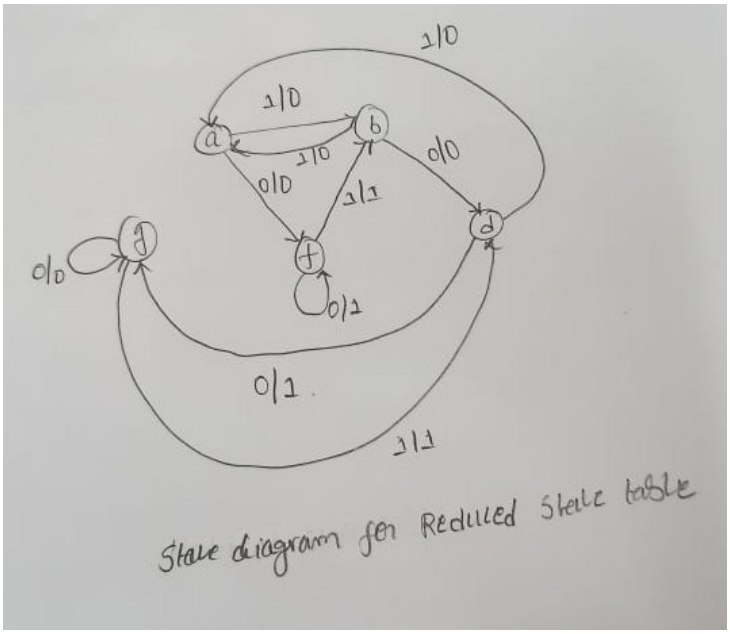


Figure 9: State Diagram Corresponding to Reduced State table

4 states ⇒ 2 flip-flops

	Present State		x	**Next State**	
	Q_1	Q_0		Q_1^+	Q_0^+
a {	0	0	0	0	0
	0	0	1	0	1
b {	0	1	0	1	1
	0	1	1	1	0
c {	1	0	0	0	0
	1	0	1	0	1
e {	1	1	0	0	0
	1	1	1	1	0

The K-map solutions for the reduced state table shown in Figures 10a and 10b

7. For the state diagram shown in Figure 11

(a) Write the output sequence for the input sequence: `001010110110110111`

(b) What is the functionality of this FSM (what function does it perform) ?

(c) Identify a flip-flop that can be used to implement above FSM without any external logical gates.

Q1 \ Q0x	00	01	11	10
0	0	0	1	1
1	0	0	1	0

Result

$F = Q0\,x + \overline{Q1}\,Q0$

(a) D_1 K-map solution

Q1 \ Q0x	00	01	11	10
0	0	1	0	1
1	0	1	0	0

Result

$F = \overline{Q0}\,x + \overline{Q1}\,Q0\,\overline{x}$

(b) D_0 K-map solution

Figure 10: K-map solutions for D_1 and D_0

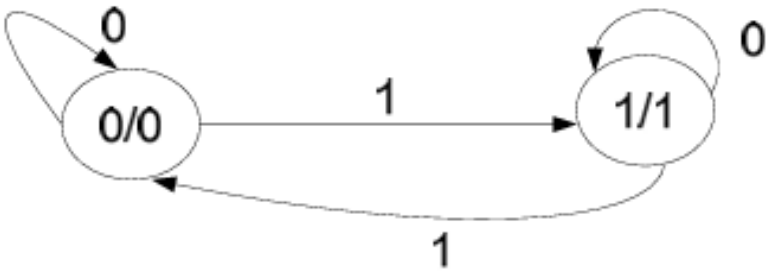


Figure 11: State diagram

Solution:

(a) **INPUT Sequence:** 0 0 1 0 1 0 1 1 0 1 1 0 1 1 0 1 1 1
OUTPUT Sequence: 0 0 1 1 0 0 1 0 0 1 0 0 1 0 0 1 0 1

(b) Note that the **OUTPUT** is 1 if the input (that has arrived till now) contains an **odd number of 1's**. So the FSM shows the functionality of a **serial odd parity indicator**.

(c) The state table is as follows:

PS	INPUT	NS	OUTPUT
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

The state table shows that:

$$\text{NS} = \text{OUTPUT} = \text{PS} \oplus \text{INPUT}$$

This is the same as the characteristic equation of a **T-Flip Flop**.

8. A combinational circuit has a static 1-hazard if, in response to any single input change and for some combination of propagation delays, the circuit output momentarily goes to 0 when it should remain a constant 1. Consider the circuit shown in Figure 12. Determine if it has a static 1-hazard by drawing a Karnaugh map. Then modify the circuit to remove the hazard.

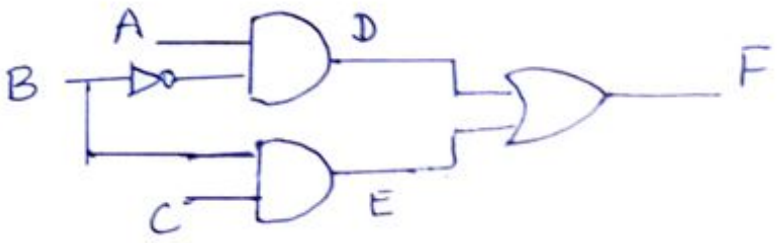


Figure 12: Circuit to check for static 1-hazard

Solution: The K-map and the modified circuit are shown in Figure 13.

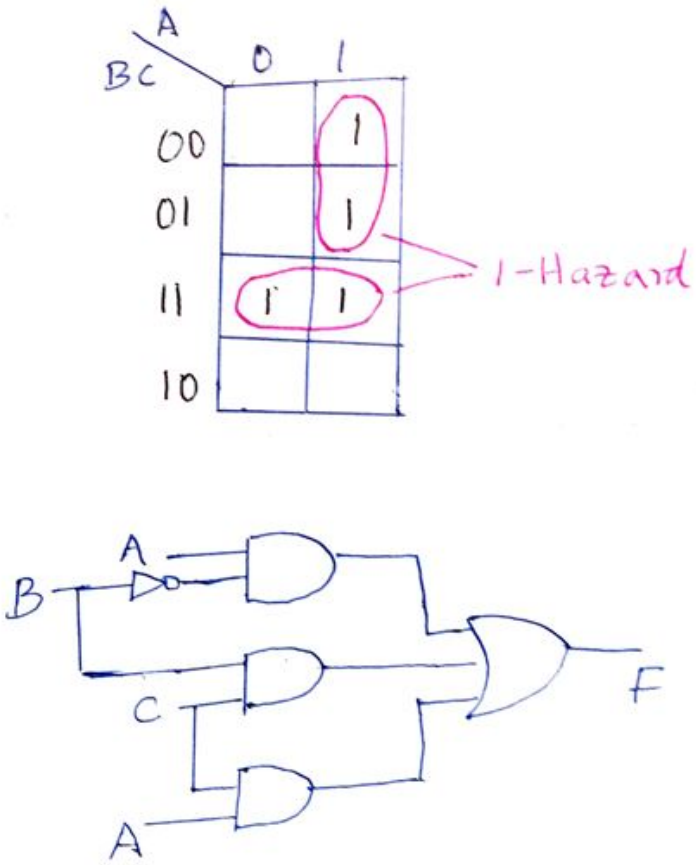


Figure 13: K-map and modified circuit without the hazard